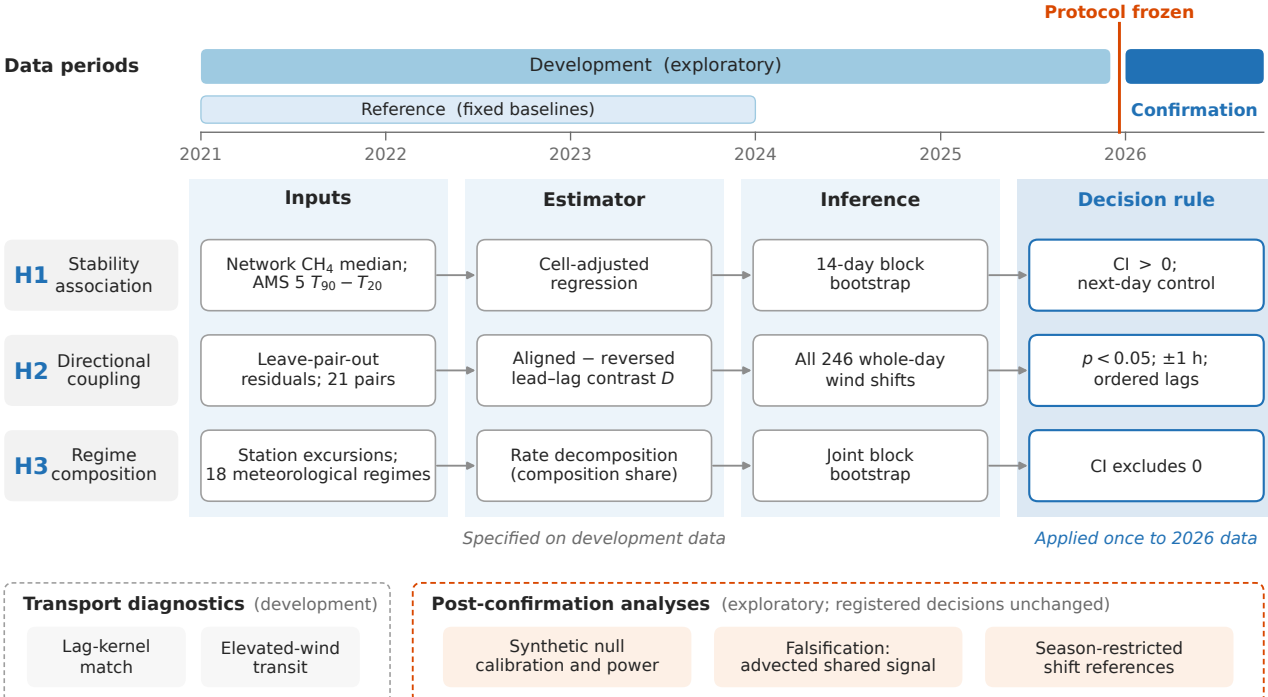
Protocol frozen
Data periods
Development (exploratory)
Reference (fixed baselines)
Confirmation
2021
2022
2023
2024
2025
2026
Inputs
Estimator
Inference
Decision rule
H1 Stability association
Network CH4 median; AMS 5 T90 − T20
Cell-adjusted regression
14-day block bootstrap
CI > 0; next-day control
H2 Directional coupling
Leave-pair-out residuals; 21 pairs
Aligned − reversed lead–lag contrast D
All 246 whole-day wind shifts
p < 0.05; ±1 h; ordered lags
H3 Regime composition
Station excursions; 18 meteorological regimes
Rate decomposition (composition share)
Joint block bootstrap
CI excludes 0
Specified on development data
Applied once to 2026 data
Transport diagnostics (development)
Lag-kernel match
Elevated-wind transit
Post-confirmation analyses (exploratory; registered decisions unchanged)
Synthetic null calibration and power
Falsification: advected shared signal
Season-restricted shift references